6 **(a)** electric and magnetic fields normal to each other B1

either charged particle enters region normal to both fields
or correct B direction w.r.t. E for zero deflection B1
for no deflection, $v = E/B$ B1 [3]

(*no credit if magnetic field region clearly not overlapping with electric field region*)

(b) (i)	$m = Bqr/v$	C1	
	$= (640 \times 10^{-3} \times 1.6 \times 10^{-19} \times 6.2 \times 10^{-2})/(9.6 \times 10^{4})$	C1	
	$= 6.61 \times 10^{-26}\,\text{kg}$	C1	
	$= (6.61 \times 10^{-26})/(1.66 \times 10^{-27})\,\text{u}$		
	$= 40\,\text{u}$	A1	[4]
(ii)	$q/m \propto 1/r$ *or* m constant <u>and</u> $q \propto 1/r$	B1	
	q/m for A is twice that for B	B1	
	ions in path A have (same mass but) twice the charge (of ions in path B)	B1	[3]